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Data Engineering and Visualization, Winter Semester 2026
Project Milestone 1
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Project Overview:

New York City never stops changing. Every day, new buildings rise, old ones are renovated, and thousands of families move in, move out, or struggle to find a home they can afford. At the same time, more than 8 million people depend on subways, buses, bikes, taxis, and streets to get to work, school, and each other. These two worlds are usually studied separately. Housing agencies look at permits, units, and violations; transportation agencies look at ridership, traffic, and crashes. But a city is a connected system: a new housing development changes who rides the subway, a crowded bus line shapes where people can afford to live, and a dangerous intersection can change the character of a neighborhood. **The most interesting questions live in the space between datasets that were never designed to talk to each other.**

The goal of this project is to navigate the complete data engineering process while **finding new, creative connections between two domains**: **housing & development** and **transportation**. You are a data detective hired by the city. You will choose **one dataset from each domain**, explore and clean them, integrate them, and build an interactive website that tells your own data story. The datasets come from NYC Open Data, and you must choose exactly one from each domain (one for housing and one for transportation):

- **Housing & Development datasets**: NYC Open Data Housing & Development catalog.
- **Transportation datasets**: NYC Open Data Transportation catalog.

Each selected dataset must contain at least 9 columns and provide at least one meaningful attribute that can be used to integrate it with the other dataset, such as time, borough, community district, ZIP code, census tract, address, latitude/longitude, or another geographic identifier. You decide the question and the angle. There is no single right answer, only well-supported, creative, and honest ones.

Dataset Catalog: A separate catalog of recommended datasets from both domains, together with their potential integration fields, will be provided. Use it as a starting point to identify datasets that may share meaningful geographic or temporal attributes before making your final selection. The catalog is **not an exclusive list**; you are encouraged to explore other datasets within the two domains, provided they meet the project requirements. If you choose a dataset outside the catalog, include a short feasibility check in your notebook covering the number of rows and columns, the relevant geographic and/or temporal attributes, the expected level of integration, and why the combination can support your research question. A large number of records alone does not guarantee a suitable dataset; the proposed integration must result in a meaningful and interpretable relationship between the two domains.

Example questions to spark your thinking:

- Do neighborhoods with the most new construction also see growth in transit use, or are they being left behind?

- Are areas with more housing complaints or violations also those with poorer transit access?
- Where is affordable housing being built, and how easily can residents reach jobs and services from there?
- Do street safety problems cluster around new developments?
- How do bike-share or taxi patterns shift in neighborhoods undergoing rapid change?

These are only starting points; the best projects come from a question nobody has asked yet.

Your Mission: Build Your Own Story

A good data story is more than a chart. Your notebook and website must make clear:

- **The question**: What did you want to know, and why does it matter to New Yorkers?
- **The data**: Which datasets did you choose, and how did you connect them? What had to be cleaned, assumed, or left out?
- **The discovery**: What did you find? What surprised you?
- **The limits**: What can your data *not* tell us? Correlation is not causation, so be honest about it.
- **The impact**: Who could use your findings, and what should they do differently?

Your work is practice for the kind of analysis that shapes real decisions. Cross-domain insights like yours can help city planners place new housing where transit can support it, help transit agencies anticipate demand in growing neighborhoods, and help housing advocates and safety initiatives identify the communities that need attention most. Even a small, well-argued finding can change how New Yorkers understand the forces shaping their own neighborhoods.

Project Team (from 4 to 6 people per group)

Each team member must propose at least 2 research questions that investigate a meaningful relationship between the two chosen datasets. Each research question must be supported by one visualization chart that directly helps answer the question. Therefore, each team member is expected to contribute 2 research questions and 2 visualizations.

For example, a research question could be: Are areas with more new residential units served by fewer transit options per capita? The corresponding visualization should clearly illustrate the relationship between residential development and transit availability.

The total number of research questions and visualizations depends on the team size:

- **4-member team:** 8 research questions + 8 visualizations
- **5-member team:** 10 research questions + 10 visualizations
- **6-member team:** 12 research questions + 12 visualizations

Clearly indicate each member's individual contributions, including their research questions, visualizations, data cleaning tasks, dataset integration, website components, and other project work.

No extensions will be granted for milestone due dates. Late submissions will not be graded.

Getting Started: Loading the Datasets

Every NYC Open Data dataset has a unique identifier in its URL (e.g., `abcd-1234`). Add the following code to your Jupyter/Colab notebook, replacing the placeholders with the identifiers of your two chosen datasets:

```python
import pandas as pd

# Replace with the dataset IDs found in each dataset's URL
housing_id = '<HOUSING_DATASET_ID>'
transport_id = '<TRANSPORTATION_DATASET_ID>'

base_url = 'https://data.cityofnewyork.us/api/views/{}/rows.csv?accessType=download'

# Load the chosen datasets
df_housing = pd.read_csv(base_url.format(housing_id), low_memory=False)
df_transport = pd.read_csv(base_url.format(transport_id), low_memory=False)

# Quick preview
df_housing.head()
df_transport.head()
```

Listing 1: Loading a Housing & Development and a Transportation dataset

```python
import pandas as pd
from urllib.parse import urlencode

dataset_id = '<DATASET_ID>'
api_url = 'https://data.cityofnewyork.us/resource/{}.csv'

# SoQL parameters: limit the size and filter the rows you need
params = {
    '$limit': 1000000,                                  # maximum number of rows to download
    '$where': "borough = 'BROOKLYN'",                   # example filter; adjust to your dataset
    # '$where': "created_date >= '2025-01-01T00:00:00'",   # example date filter
    '$order': ':id'                                     # stable order so your sample is repeatable
}

df = pd.read_csv(api_url.format(dataset_id) + '?' + urlencode(params), low_memory=False)
df.head()
```

Listing 2: Loading a filtered sample of a large dataset

Note: Some datasets are very large (millions of rows) and may crash your notebook if downloaded in full. For these, use a filtered sample. As in Listing 2, but you must justify this choice in your notebook. The API returns only 1,000 rows if you do not set `$limit`. Column names from the API are lowercase with underscores and may differ from the downloaded CSV, so check `df. columns`.

Milestone 1: Cleaning, Integration, and Visualization

In this milestone, you will:

- **Explore the Data**: Use descriptive statistics and initial plots to understand each dataset's structure, issues, and patterns.
- **Clean the Data (Pre-Integration)**:
  - Handle missing values (justify drop vs. impute).
  - Detect and address outliers (e.g., IQR, domain rules).
  - Check inconsistent data
  - Standardize formats (dates, strings, categories, and geographic identifiers).

  - Remove duplicates.

- **Integrate the Datasets**: The two datasets share no ready-made key, so finding a meaningful link is part of the challenge. Possible bridges include **time** (year, month), **administrative geography** (borough, community district, ZIP code, census tract), or **spatial proximity** (latitude/longitude, spatial joins, distance to the nearest station or stop). Document your integration steps, justify your choice of join key and granularity, and explain any aggregation needed to align the datasets.

  Important: Teams do not need to match individual records from the two datasets. If the datasets have different levels of detail or very different numbers of records, they may aggregate the data to a common geographic or temporal level (such as ZIP code by year) before integration. Teams must explain why the chosen level is appropriate for their research question.

- **Post-Integration Cleaning** *(Required)*: After joining, resolve:
  - New missing values from joins.
  - Inconsistent or redundant columns.
  - Data type mismatches.

- **Build an Interactive Website:** Create a fully interactive data visualization website using **Dash (Python/Plotly)** or any web framework such as React with Plotly.js, Flask, or Django. The website should provide an engaging and dynamic way for users to explore your integrated dataset and follow your story.

  The website must include the following structure and functionality:
  - Multiple **dropdown filters** (e.g., Borough, Year, a category from the housing dataset, a category from the transportation dataset) allowing users to dynamically filter data.
  - A **search mode**, where users can type queries (e.g., "Brooklyn 2022 new buildings") to automatically apply filters.
  - A central **"Generate Report" button** that, when clicked, dynamically updates all visualizations based on the selected filters or search terms.
  - Visualizations should include a variety of chart types, such as bar charts, line charts, heatmaps, maps, or scatter plots, and must offer interactivity (hover, zoom, or filter updates).
  - At least one visualization must show the **relationship between the two domains** (e.g., a map overlaying both datasets or a scatter plot of a housing metric against a transportation metric).
  - A short **story section** on the website presenting your question, key findings, limitations, and the benefit to NYC.
  - Ensure that all components respond in real time to user interactions and that the website layout is user-friendly and visually consistent.
  - Host and test the website using a free deployment platform such as **Vercel**, **Render**, or **Heroku**, and confirm full functionality before submission.

The Grades Will Be Given For:

- **Implementation**:
  - Thorough exploratory data analysis (EDA) with relevant statistics and visualizations.
  - Effective **pre- and post-integration cleaning** (e.g., appropriate handling of missing values, outliers, and inconsistencies).
  - Quality of integration: a justified join strategy and clear added value from combining the two domains.
  - Relevance and interactivity of website visualizations (e.g., do dropdowns + **Generate Report button** drive meaningful updates?).

  - Creativity and originality of research questions (e.g., avoid simplistic counts; aim for unexpected connections, spatial trends, or predictive patterns).
  - Quality of the story: a clear question, honest limitations, and a convincing "so what?" for NYC.
- **Descriptive Markdown Cells**: Document exploration, cleaning, and integration steps. Justify decisions (e.g., why drop vs. impute nulls?), discuss alternatives considered, and explain how insights were reached.
- **Clean Code**: Use descriptive variable names, modular functions, and clear structure.
- **Code Comments**: Comment on complex logic (e.g., outlier detection algorithms, integration joins).
- **Deductions** for:
  - Poor contribution documentation.
  - Inappropriate plot choices or unclear labels.
  - Messy or uncommented code.
  - Incorrect or trivial conclusions, or claims of causation unsupported by the data.
  - Missing **"Generate Report" button** or post-integration cleaning.

Deliverables

- A **Jupyter/Colab notebook** containing:
  - Dataset overview for both chosen datasets (e.g., number of rows, number of columns, relevant geographic and temporal attributes, known issues) and the reasoning behind the selection. For datasets with very different sizes or granularities, explain how the data will be aggregated or transformed to enable a meaningful integration.
  - Detailed steps for **EDA**, **pre-integration cleaning**, **integration**, and **post-integration cleaning**.
  - Visualizations supporting data understanding and validation of cleaning steps.
  - Prompt used in any step in the project and Links for chosen datasets.
- A **functional website** demonstrating interactive exploration, reporting, and storytelling, including:
  - Dropdown-based filtering, search mode, and "Generate Report" interactivity.
  - Deployed version (e.g., on Vercel) and full source code in a GitHub repository.
  - A **README file** that includes setup steps, deployment instructions, and a short description of each team member's contribution.

Submission Instructions

All project code, notebooks, and website files must be hosted on a **GitHub repository**. Each team must submit **one email** to ICS503Data@gmail.com that includes:

- The **IDs** of all team members.
- The **GitHub repository link**.
- The **deployment link** of the website.
- The **links to the datasets** used in the project.

Make sure the repository contains all required files before the deadline. Changes made after the deadline (**9 November 2026**) will not be considered.